HygionX - Bridging Artificial Intelligence & Explainable Medicine

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ABSTRACT

- This project presents HygionX, a Flask-based AI-powered medical triage web application that provides intelligent and structured healthcare guidance. It allows users to describe symptoms in natural language and performs symptom extraction, follow-up questioning, and severity assessment to generate accurate triage outcomes. The system uses Natural Language Processing, rule-based logic, and optional machine learning models to classify risk levels such as low, moderate, high, or emergency, while ensuring explainability through clear reasoning, confidence levels, and recommended actions. HygionX maintains user history and session continuity by securely storing conversations, symptoms, and triage results, enabling personalized responses. It also includes features like medical knowledge lookup, health timeline tracking, and user profile management. Overall, the project highlights the potential of AI-driven triage systems in improving accessible, reliable, and user-centric digital healthcare.

INTRODUCTION

- Healthcare access is often slow and inconsistent, making it difficult for patients to receive timely guidance for common symptoms.
- Traditional support systems lack personalization, leading to generic advice and limited continuity of care.
- The project **HygionX** provides an AI-driven solution that uses NLP, ML, and Explainability to analyze symptoms and deliver accurate, transparent medical guidance.
- This approach makes healthcare faster, personalized, and more accessible, improving telemedicine and overall patient engagement.
- The system supports personalized health insights by leveraging patient history and previous interactions to deliver more relevant and context-aware recommendations

PROBLEM STATEMENT

- Existing healthcare chatbots do not consider patient medical history, leading to incomplete or generic responses.
- Traditional support systems lack personalization, leading to generic advice and limited continuity of care.
- Most systems provide answers without explainability, creating trust issues for users.
- Inadequate integration of user health history leads to less personalized responses.
- Most of the existing chatbots donot predict the triage efficiently, leading to a gap in medical analysis
- Therefore, there is a need for a chatbot that offers personalized, explainable, and history-aware medical support, bridging the gaps in existing solutions.

EXISTING SYSTEM

- Current healthcare chatbots such as Babylon Health, Buoy Health, and Ada Health mainly focus on general symptom checking, while platforms like Wysa, Woebot, and Replika provide emotional or mental-health support. Similarly, bots like Florence and Molly assist users with medication reminders and chronic condition management, and hospital-specific chatbots from institutions like the Mayo Clinic and Cleaveland Clinic offer basic medical guidance for their patients. However, these systems largely rely on static question-answer flows and do not integrate a user's complete medical history while generating responses. Most existing chatbots operate reactively, responding only when asked, and lack features such as multilingual support, deep personalization, or transparent explainability of their answers. As a result they often struggle with complex or rare medical cases, and fail to establish the trust due to limited reasoning visibility and weak integration with patient records or healthcare systems.

DISADVANTAGES

- Uses basic symptom matching without considering the deeper medical context.
- Limited adaptation to individual patient differences.
- No continuous patient profile or long-term memory.
- Reasoning behind responses is not clearly explained.
- Accuracy drops for complex or rare medical cases.
- Weak integration with hospitals or medical systems.

PROPOSED SYSTEM

- The proposed system introduces **HygionX,** an AI-powered healthcare chatbot that combines user medical history with real-time symptom inputs to deliver personalized and reliable guidance. It uses HuggingFace transformers like MiniLM for sentence transformations and SciSpacy NER transformer for efficient symptom recognition. And the system acts by predefined secure rules with preprocessed data, return a JSON output. A secure MySQL database maintains user profiles and past records, enabling continuity across sessions. The system incorporates to highlight important factors, reasoning, and confidence levels behind every prediction, improving trasparency and trust. HygionX also aims to support accurate disease prediction using pre trained models, update user history after each session, and also trigger a “consult doctor” recommendation when confidence falls below a threshold, ensuring safe and responsible usage.

ADVANTAGES

- Personalized medical guidance
- History-aware responses
- Explainable and transparent outputs
- Higher accuracy in answers
- Safe, confidence-based suggestions
- Continuous user data updates
- Scope of updation of user data
- Secured user data in efficient databases
- Firebase Authentication

SYSTEM REQUIREMENTS

Software Requirements:

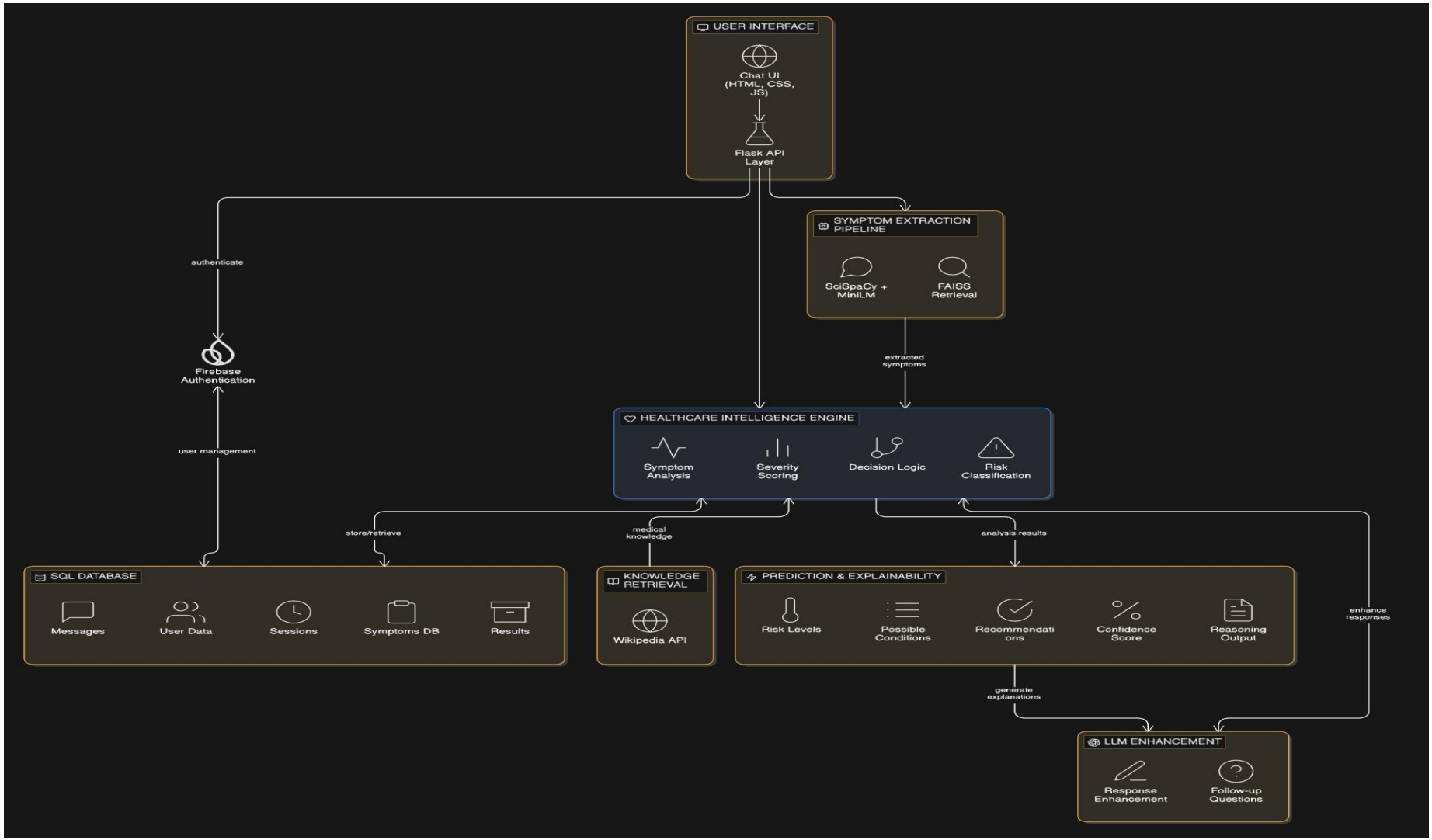
Operating System	: Windows 10/11
Browser	: Google Chrome/ Firefox
Programming language	: Python
Frontend	: HTML, CSS, JS
Backend Framework	: Flask (REST API)
Databases	: MySQL - user records
Authentication	: Firebase
NLP/ML Libraries	: PyTorch, HuggingFace Transformers
Development Tools	: VS Code, Google Colab, Jupyter Notebook
Version Control	: Git, GitHub

SYSTEM REQUIREMENTS

Hardware Requirements:

Processor : Intel Core 2 Duo / i3 Processor (1.8 GHz or higher)
RAM : 4GB minimum (8GB recommended)
Hard disk : 20 GB or more
Display : Standard HD Monitor
Keyboard : Standard 104-keys keyboard
Mouse : Optical mouse
Internet : Stable broadband connection

SYSTEM ARCHITECTURE



SYSTEM ARCHITECTURE

- The system follows a modular client-server architecture, separating the user interface, backend processing, and data storage layers to ensure scalability and maintainability.
- The application provides a web-based chat interface for user interaction, with secure authentication handled through Firebase Authentication.
- A Flask-based backend API manages request handling, session management, and communication between system components, enabling smooth data flow.
- The core Healthcare Intelligence Engine processes user inputs using NLP and ML techniques, including symptom extraction and triage logic for severity assessment and risk classification.
- User data, chat history, detected symptoms, and triage results are stored in a SQL database, while external services such as knowledge retrieval APIs and LLM modules enhance system functionality and responses.

MODULES

- User Module
- Admin Module
- Authentication & User Management Module
- Conversation & Session Management Module
- Knowledge Retrieval Module
- Healthcare Intelligence Engine Module
- Database & Infrastructure Module

MODULES

- **User Module**:

The User Module allows patients to interact with the HygionX system through a web-based chat interface. Users can register and login securely to access the platform. They can enter symptoms in natural language and receive AI-generated healthcare guidance. The module also displays responses with explanations and suggested actions. Additionally, users can view past interactions and manage basic profile information. This ensures a smooth and personalized user experience.

MODULES

- **Admin Module:**

The Admin Module is responsible for managing and monitoring the overall system operations. It allows administrators to oversee user activity and system performance. The module helps in maintaining datasets, configurations, and backend services. It ensures that the system runs efficiently without disruptions. It also supports monitoring of stored medical data and system outputs. This contributes to reliability and proper system maintenance.

MODULES

- **Authentication & User Management Module**:

This module handles secure user authentication and account management. It enables users to register and login using a secure authentication system. User details such as name, age, gender, and known conditions are stored safely. The module ensures proper access control and protects user data. It also supports personalized experiences based on user profiles. This improves both security and usability of the platform.

MODULES

- **Conversation & Session Management Module**

This module manages the entire interaction flow between the user and the system. It maintains session continuity by storing chat history and context. Each message is recorded and linked to a session for better tracking. The module keeps track of detected symptoms across multiple messages. It ensures that follow-up questions are relevant and meaningful. This improves the accuracy and consistency of responses.

MODULES

- **Knowledge Retrieval Module**

The Knowledge Retrieval Module provides additional medical information to users. It retrieves disease-related details from external sources such as medical databases or APIs. The module presents this information in a simplified and user-friendly format. It helps users understand conditions beyond just predictions. It also enhances the overall learning experience for the user. This makes the system more informative and supportive.

MODULES

- **Healthcare Intelligence Engine:**

This is the core AI-driven component of the HygionX system. It processes user input using NLP techniques to extract relevant symptoms. The module analyzes severity levels and identifies possible conditions. It generates risk classifications such as low, moderate, or high. It also provides recommended actions and basic explanations for the results. This ensures accurate, explainable, and useful healthcare guidance.

MODULES

- **Database & Infrastructure Module:**

This module is responsible for handling data storage and system infrastructure. It stores user profiles, chat history, detected symptoms, and triage results. The system uses a structured database to maintain consistency and reliability. It supports efficient retrieval of past data for continuity. The architecture is designed to handle failures gracefully without breaking the system. This ensures scalability, stability, and long-term usability.

UML DIAGRAMS

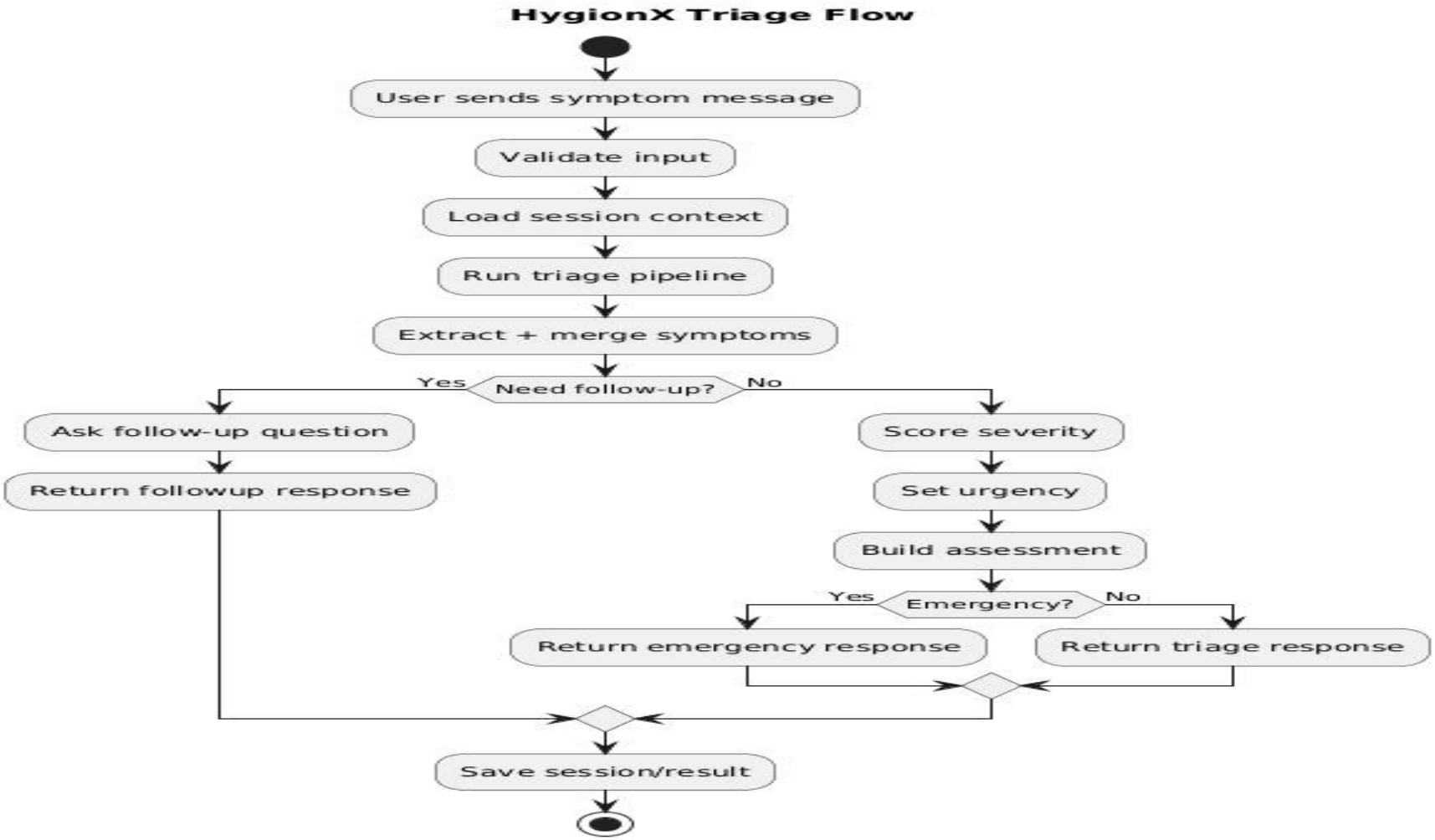


Fig: Activity Diagram

UML DIAGRAMS

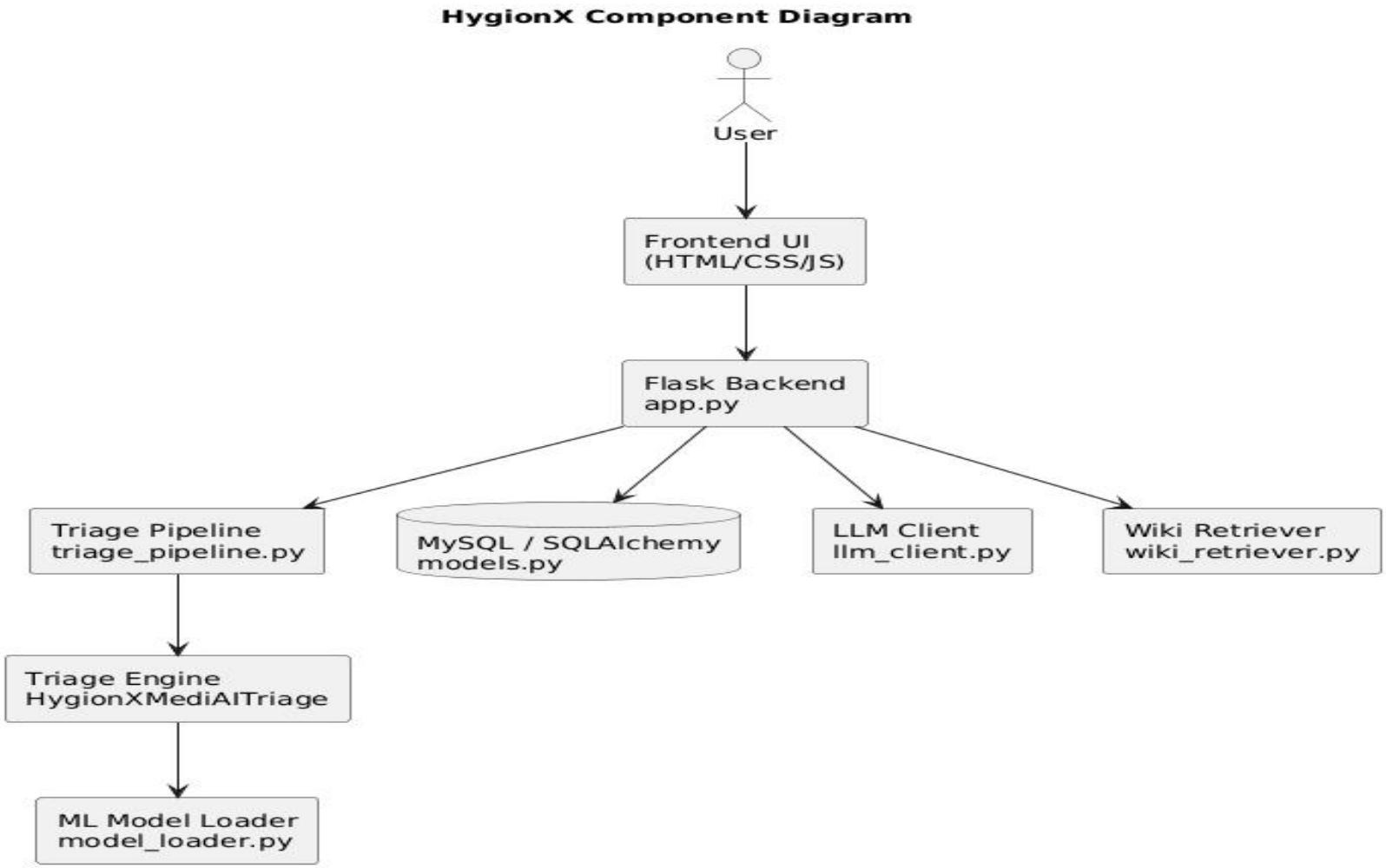


Fig: Component Diagram

UML DIAGRAMS

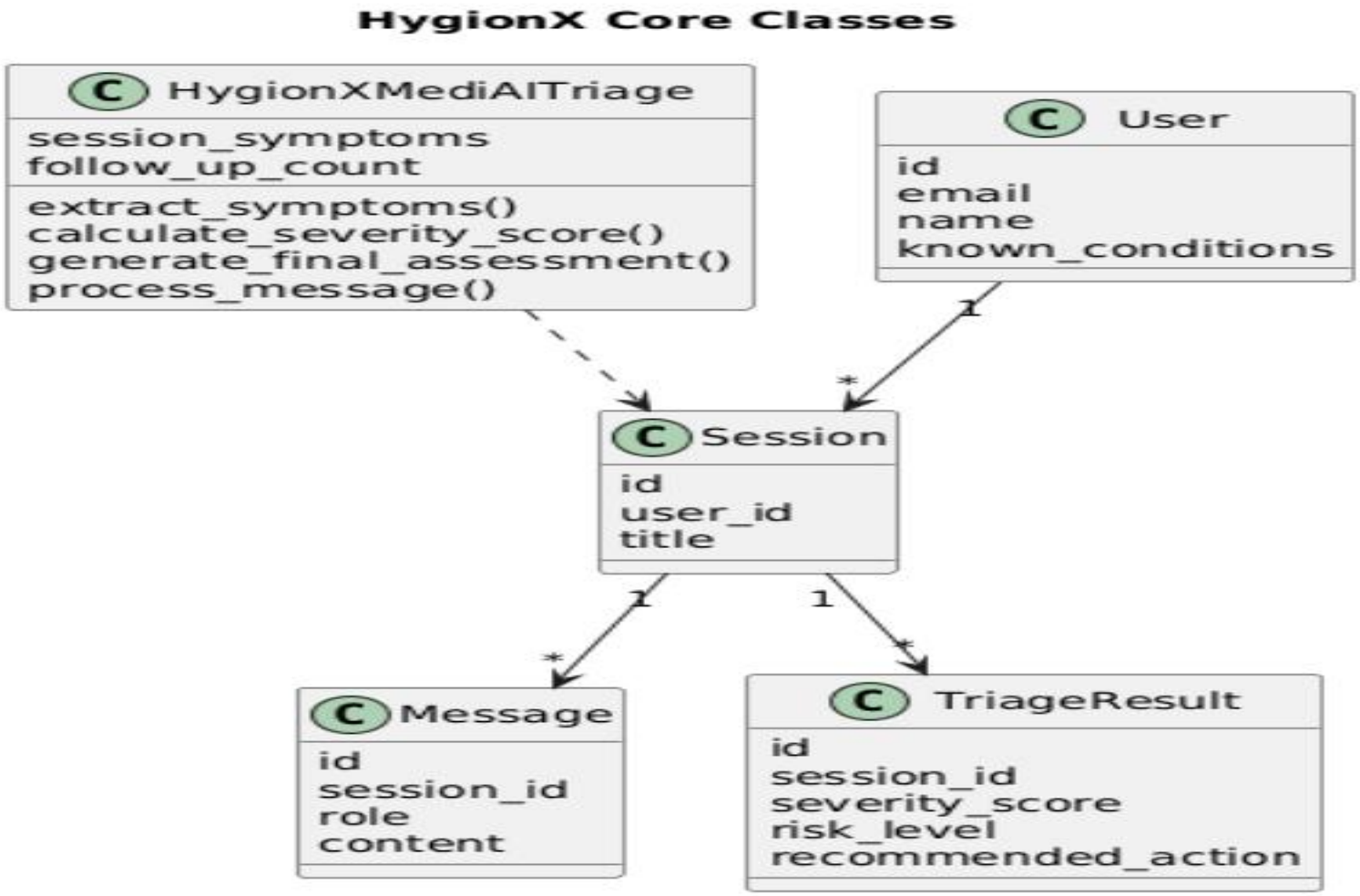


Fig: Class Diagram

UML DIAGRAMS

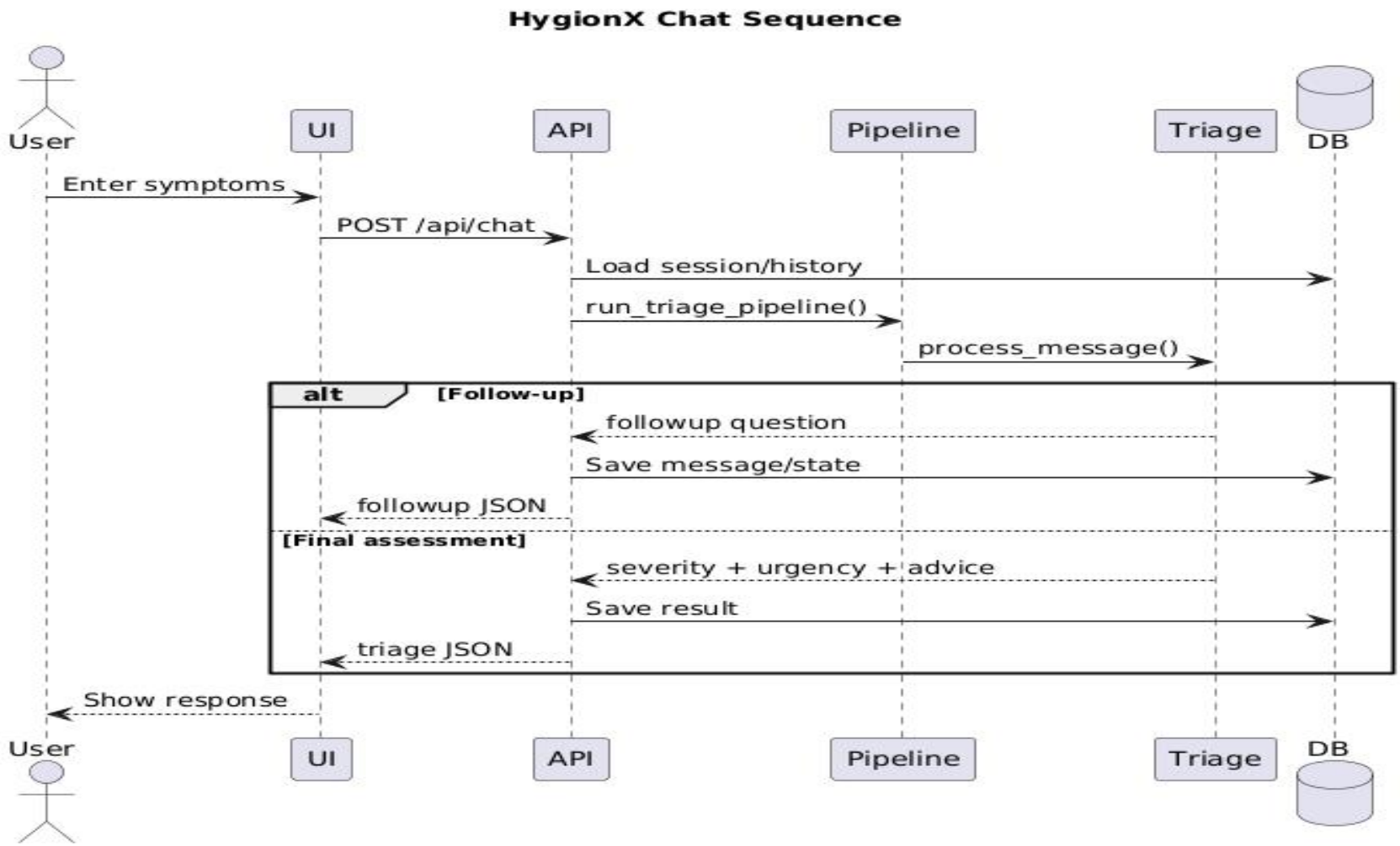
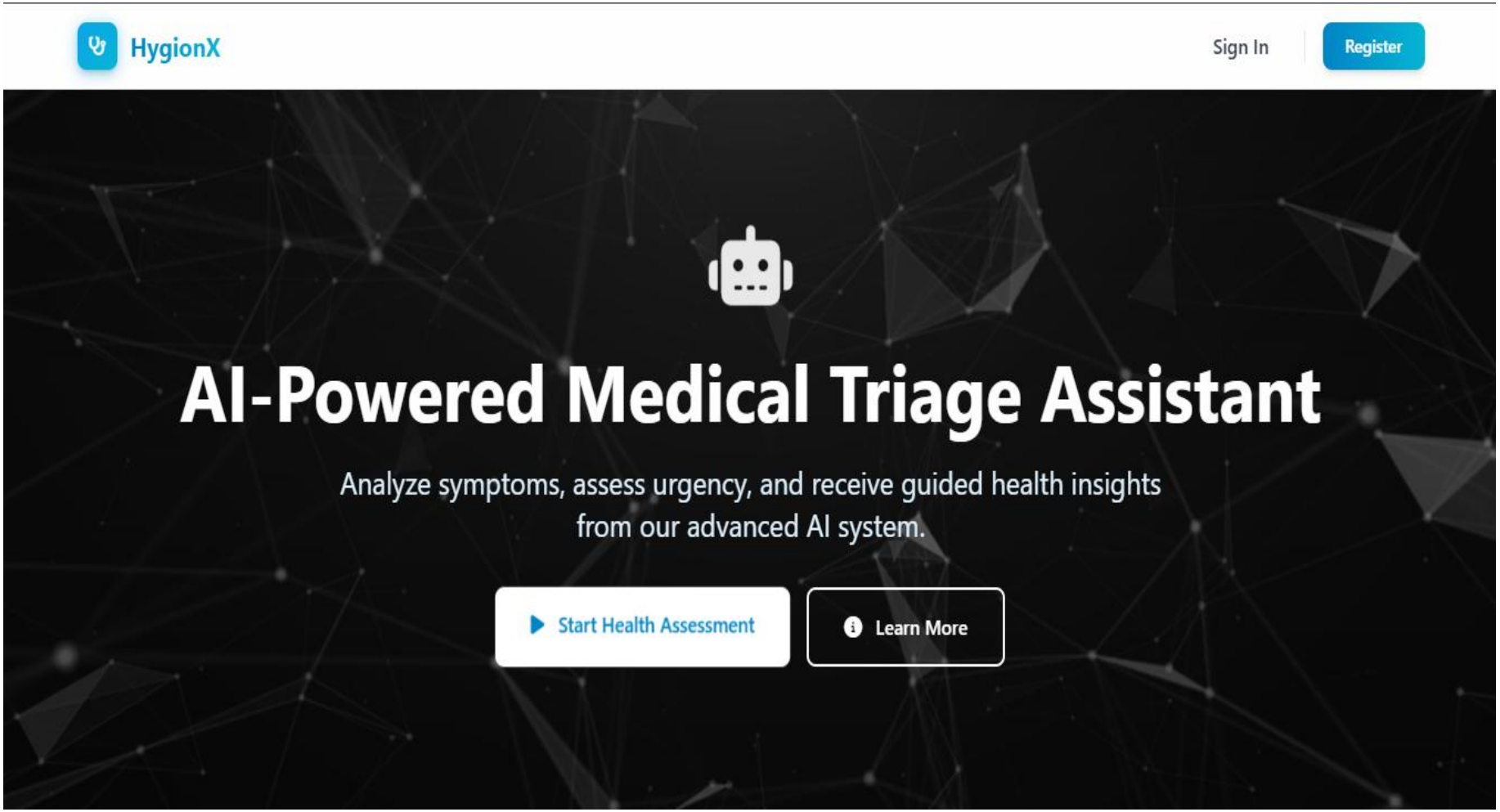


Fig: Sequence Diagram

OUTPUT

Landing Page



OUTPUT

Health Assistant

Health Assistant
AI-powered symptom analysis and medical guidance

Low Risk | Session Summary | U

- Health Assistant
- Health Timeline
- Medical Encyclopedia
- Profile
- Settings
- Help
- About
- Logout

AI Assessment

Assessment

Based on the symptoms detected in this session: fever. This single symptom may indicate various medical conditions ranging from mild to serious depending on severity and duration. The severity score of 2/10 was assigned based on the number and type of symptoms present. These symptoms can often be managed at home with monitoring. Professional medical evaluation is recommended for proper diagnosis and treatment.

Urgency Level

Low Risk

Recommended Action

Monitor symptoms at home with rest and hydration. Over-the-counter medications may help if used according to label instructions. Seek medical care if symptoms persist or worsen.

Possible Conditions

Flu	47%
COVID-19	33%
Common Cold	20%

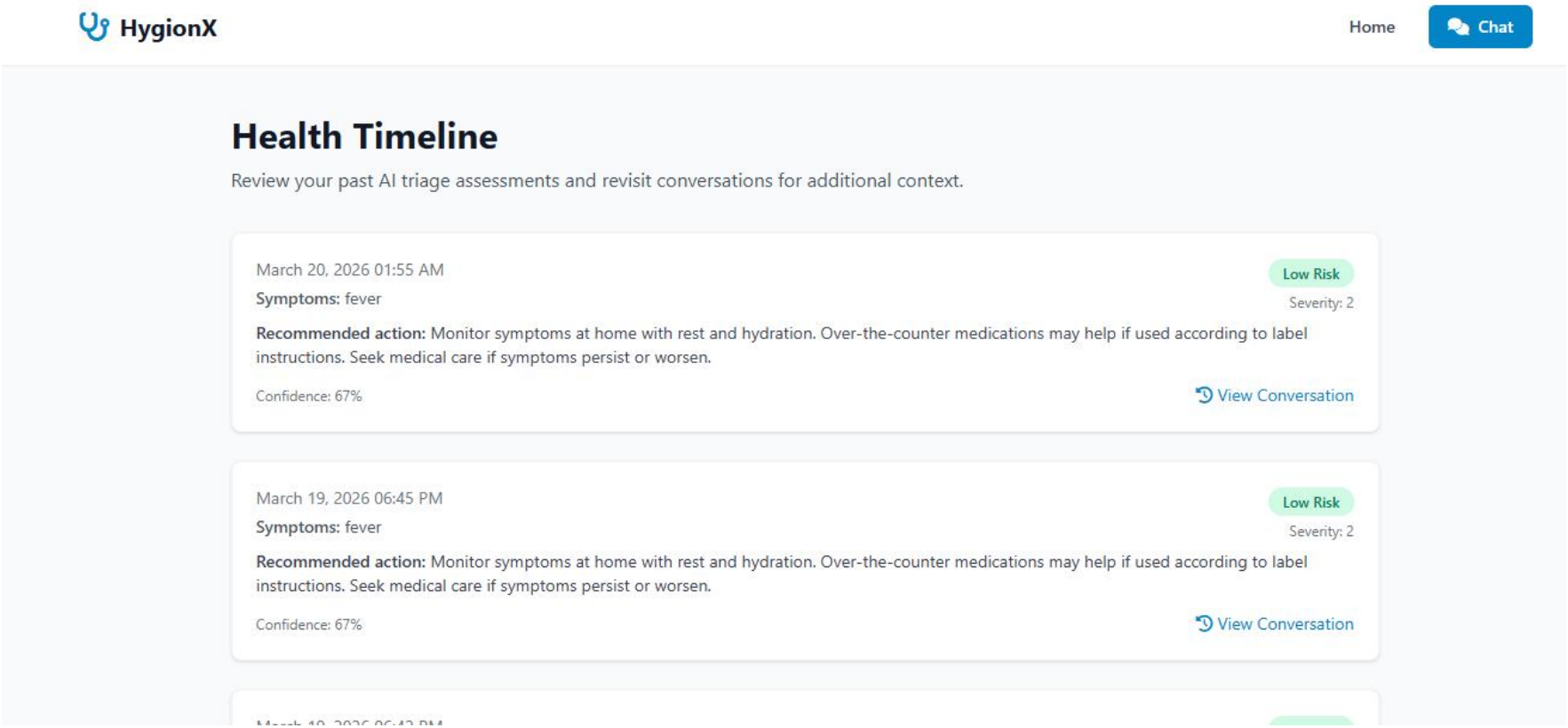
This AI provides guidance only and is not a substitute for professional medical diagnosis.

AI Assistant • Just now

Describe your symptoms in detail... | Send

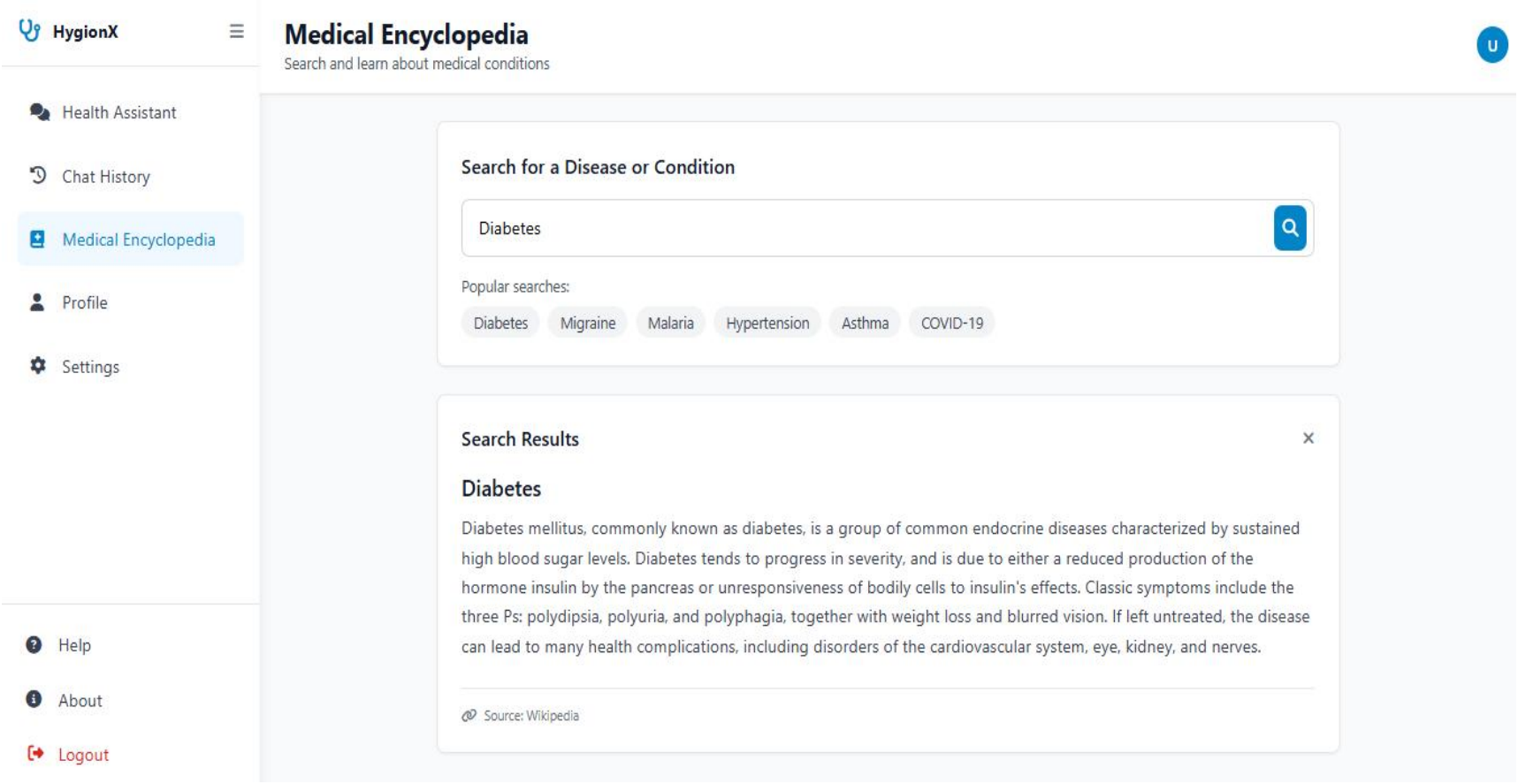
OUTPUT

Health Timeline



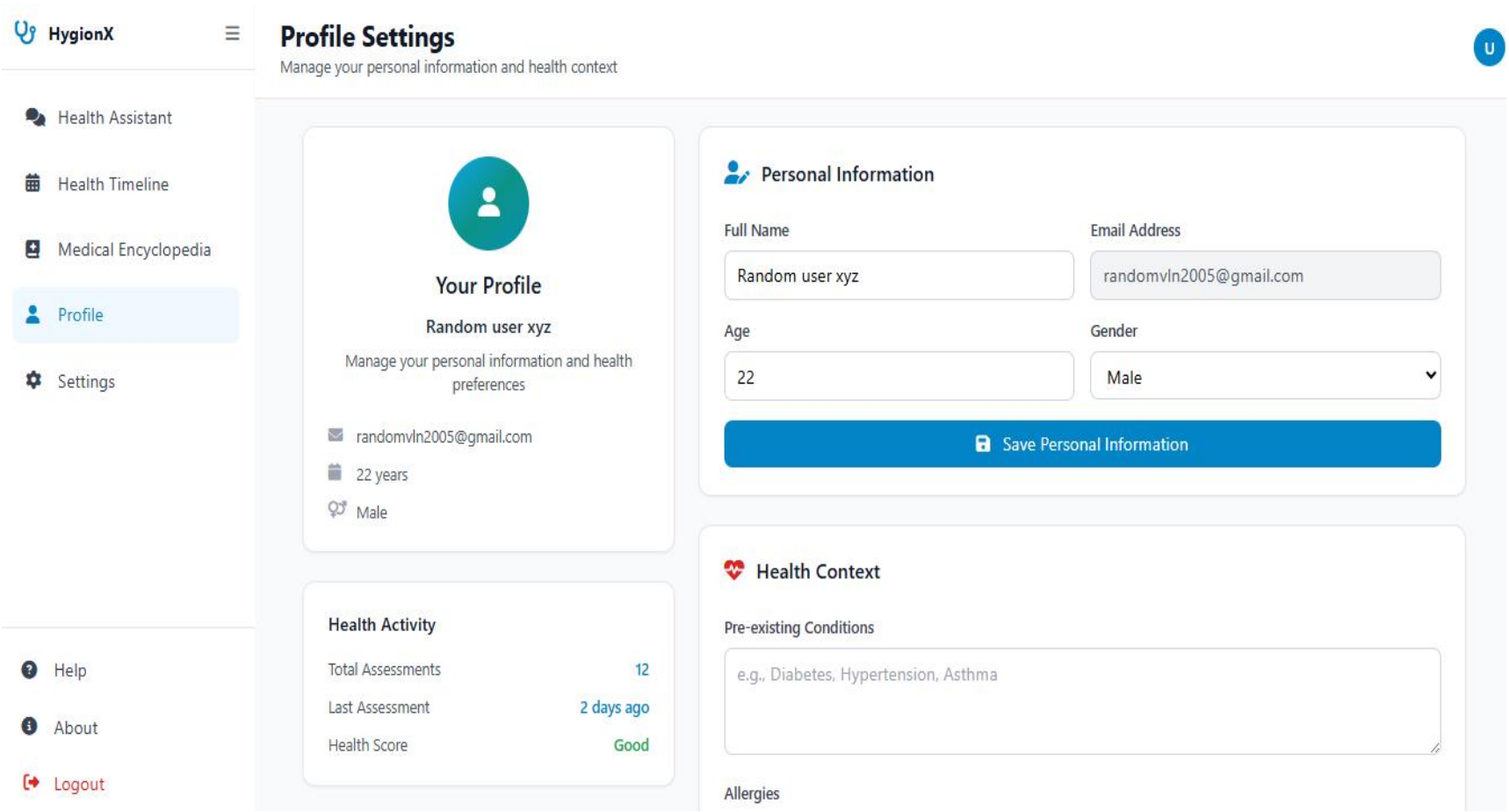
OUTPUT

Medical Encyclopedia



OUTPUT

User Profile



FUTURE SCOPE

- The system can be enhanced by integrating voice-based interaction to allow users to describe symptoms through speech, improving accessibility and ease of use.
- Advanced machine learning models can be incorporated to improve prediction accuracy and support more complex medical conditions and personalized recommendations.
- Integration with wearable devices and real-time health monitoring systems can enable continuous tracking of patient health data.
- The platform can be extended to include direct doctor consultation, appointment booking, and telemedicine support for complete healthcare assistance.

CONCLUSION

- HygionX demonstrates the effective use of AI, NLP, and ML in providing smart healthcare assistance.
- It delivers personalized and history-aware medical guidance, improving accuracy over existing systems.
- The system ensures explainable results, increasing user trust and transparency.
- Its triage-based approach helps users understand severity, urgency, and next steps clearly.
- Overall, it acts as a reliable first-level healthcare support system, enhancing accessibility and telemedicine.

Thank you!